

A Hierarchical Shell-Manifold Theory: Foundational Paper

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Abstract

Hierarchical Shell-Manifold Theory (HSMT) is a speculative geometric framework in which spacetime is organised as dimensionally nested, spatially immanent grades labelled by an integer $N \in \mathbb{Z}$. A Dirac-type operator organises the grades. Residual coupling is a Gaussian overlap whose width is constrained by the demand that large-scale geometry appear statistically isotropic and homogeneous. Time on a grade is not a coordinate of the arena. It is the non-stationarity left after residual coupling is projected onto that grade.

This document is the foundational paper of the theory. It supersedes the May 2026 draft. Candidate geometric objects from that draft are retained. Completed-unification claims, uniqueness at $\alpha = \pi + G$, and numerical Standard Model or cosmological fits are not. The working content is an associative reference model on $L^2(\mathbb{R}^3 \times \mathbb{Z}, \mathbb{C}^2)$: a self-adjoint operator, an isolation window for grade $N = 4$, an exact projected equation for the leak, and a projected heat kernel whose spectral-dimension limits are not moved by the leak. An arrow of time is not derived.

1 Status of this paper

The May 2026 manuscript introduced the geometric objects of HSMT and attached to them a two-pillar unification programme and a large phenomenological list. The July 2026 note kept the objects and refused the stronger claims. The September 2026 reference model supplied the first operator-level statements that can be called results rather than slogans.

This paper is the May foundational document rewritten under those later rules. Where May and September disagree, September governs. The May file is a historical source of candidates, not the official theory.

What is retained from May: integer grades, an abstract hierarchical coordinate, a Gaussian residual kernel, a Dirac-type organiser, a measure intended to carry dimensional flow, and the strategy of reading low-energy structure from well-defined overlaps.

What is not retained as results: uniqueness of the theory, exact octonionic eigenfunction status for an entire tower, and every numerical May prediction (gauge couplings, Higgs mass, proton lifetime, relic density, superconducting T_c , inflationary pair, global χ^2). Those statements are withdrawn. They may be recomputed only after the operator, the measure, and the

projection are specified well enough to make the integrals exist.

2 Core objects

Definition 1 (Graded arena).

$$\mathcal{M} = \bigsqcup_{N \in \mathbb{Z}} \mathcal{M}_N.$$

The grades are dimensionally nested and spatially immanent. Lower grades are constitutively present in higher grades as geometric foundation. They are not thin shells at separate locations. No global centre is required.

The integer N labels the grade. A K-theory or Chern-number reading of N is optional and unproved. The observed macroscopic world is associated with a distinguished grade written $N = 4$. That assignment is a modelling choice motivated by an effective dimension near four. It is not a dynamical selection theorem.

Definition 2 (Hierarchical coordinate). *The coordinate ρ measures separation between grades. It is not cosmological distance, redshift, or an observational distortion.*

Ansatz 1 (Unit map).

$$\rho_N = N - 4, \quad N \in \mathbb{Z}.$$

Adjacent grades are at hierarchical distance 1. Other constant spacings rescale σ_0 . Logarithmic and spectral maps remain legal and are not used here.

Definition 3 (Residual kernel).

$$w(\rho) = \frac{1}{\sqrt{2\pi} \sigma_0} \exp\left(-\frac{\rho^2}{2\sigma_0^2}\right).$$

The kernel never vanishes. Residual coupling is always present. The width σ_0 is constrained by a continuum demand: mixing must be small enough that large-scale geometry can appear statistically isotropic, homogeneous, and effectively smooth, and large enough that projected states are not frozen.

On the unit lattice,

$$r(k; \sigma_0) = \exp(-k^2/(2\sigma_0^2)), \quad p(k; \sigma_0) = \frac{r(k; \sigma_0)}{\sum_m r(m; \sigma_0)}.$$

A first window with nearest-neighbour dominance is

$$0.30 \lesssim \sigma_0 \lesssim 0.40.$$

The May candidate $\sigma_0 = \sqrt{2}/4 \approx 0.354$ lies inside that window: $p(0) \approx 0.965$, $p(1) \approx 0.018$, $r(2) \approx 10^{-7}$. It is not selected uniquely by the window.

The table weight $r(1)$ is not a matrix element. The working hop is

$$t = t_0 \sqrt{r(1; \sigma_0)} = t_0 \exp(-1/(4\sigma_0^2)). \quad (1)$$

At $\sigma_0 = \sqrt{2}/4$, $t \approx 0.135 t_0$. Every formula in t^2/v^2 uses (1).

Ansatz 2 (Multifractal weight). *A measure on the grades is intended to carry dimensional flow and content flow from deeper grades into a macroscopic continuum. The older schematic*

$$d_N(\ell) = d_\infty + \alpha(\ell_0/\ell)^\gamma$$

is not used as a working formula. Observed flow is carried by a running spatial order on grade 4, Section 7.

3 Projectors

May used the same Gaussian as a coherent-state integral

$$G_N = \int w(\rho) |\psi_N(\rho)\rangle \langle \psi_N(\rho)| d\mu(\rho)$$

and inserted G_N into a spectral action. That construction is a modelling device only.

Definition 4 (Grade projector). Φ_N is the orthogonal projector onto the grade- N summand of the working Hilbert space. If ψ is already supported on grade N , then $\Phi_N \psi = \psi$. Residual coupling is a property of the organiser $\mathcal{D}$, not of Φ_N .

Ansatz 3 (Layer projection). *Observable 3+1 physics is the effective description after projection onto $N = 4$. Other grades are not elsewhere in space. They are hierarchically present and weakly coupled through $w(\rho)$.*

This does not prove that $N = 4$ is selected, that Lorentzian signature or chirality comes from Φ_4 , or that the projected theory reproduces local quantum field theory to experimental accuracy.

4 Working operator

Octonionic bi-multiplication is deferred. The foundation uses associative fibres.

The working Hilbert space is

$$\mathcal{H} = L^2(\mathbb{R}^3 \times \mathbb{Z}, \mathbb{C}^2).$$

Spatial derivation and hierarchical hopping act on one two-component fibre.

Ansatz 4 (Working operator).

$$(\mathcal{D}\psi)_N = D_{\text{sp}}(N)\psi_N + t\sigma^1\psi_{N+1} + t\sigma^1\psi_{N-1} + W_N\sigma^2\psi_N + m\sigma^3\psi_N, \quad (2)$$

with hop (1) and odd on-site profile

$$W_N = \kappa \tanh(\alpha(N - 4)).$$

The gap scale on the first neighbours is

$$v := |W_5| = |\kappa \tanh \alpha|.$$

Second-neighbour hops are omitted inside the σ_0 window.

The default spatial operator on the distinguished grade is the Weyl Hamiltonian

$$D_{\text{sp}}(4) = -i\sigma \cdot \nabla. \quad (3)$$

There is no coordinate time on $\mathbb{R}^3$. Grade-dependent and scale-dependent replacements for $D_{\text{sp}}(N)$ are introduced in Section 7.

May's continuum parent

$$\mathcal{D}_\rho = i\sigma^1 \frac{d}{d\rho} + \frac{A}{2}(L_{u(\rho)} + R_{u(\rho)})\sigma^2 + m_0\sigma^3$$

remains a formal expression. If the multiplication algebra is $\mathbb{C}$ or $\mathbb{H}$, it is an ordinary Dirac-type operator once potentials are specified. If the algebra is $\mathbb{O}$, essential self-adjointness and the status of a spectral triple are open. They are not used as input here.

Proposition 1 (Self-adjointness). *On the dense domain of smooth compactly supported spinors, occupying finitely many grades, the operator (2) with $D_{\text{sp}}(N)$ a Fourier multiplier of order γ_N bounded and bounded away from zero is essentially self-adjoint on $L^2(\mathbb{R}^3 \times \mathbb{Z}, \mathbb{C}^2)$.*

Each slice operator is essentially self-adjoint. Hierarchical terms are bounded. Kato–Rellich applies.

On grade 4, with (3) and $W_4 = 0$,

$$\Phi_4 \mathcal{D} \Phi_4 = -i\sigma \cdot \nabla + m\sigma^3.$$

Stone's theorem supplies a unitary group

$$U_4(t) = \exp(-it(-i\sigma \cdot \nabla + m\sigma^3)). \quad (4)$$

The parameter t is the effective time of the grade. It is not ρ and not a coordinate of $\mathbb{R}^3$.

What this does not give: a four-component Dirac fermion, a Lorentzian metric tensor as a derived object, chirality as a hierarchical output, or an arrow.

5 Time

The arena has no primitive time. Increasing N is not cosmological time.

Ansatz 5 (Incomplete confinement). *A state may satisfy $\Phi_N \psi = \psi$ and still fail to remain on that grade, because*

$$\Phi_N \mathcal{D}(1 - \Phi_N) \neq 0.$$

The image of that operator is a threshold amplitude: semi-presence of the neighbours, not residence on them.

The un-eliminated split is

$$\partial_{t_N}\psi_N = \Phi_N\mathcal{D}\Phi_N\psi_N + \Phi_N\mathcal{D}(1 - \Phi_N)\psi. \quad (5)$$

After the neighbour amplitude is solved out, and with only $N = 3, 5$ retained, the exact projected law is

$$i\partial_t\psi_4 = \Phi_4\mathcal{D}\Phi_4\psi_4 + \int_0^t K(t-s)\psi_4(s)ds + \xi(t), \quad (6)$$

$$K(\tau) = -it^2(e^{-iD_3\tau} + e^{-iD_5\tau}),$$

$$\xi(t) = t\sigma^1 e^{-iD_3t}\chi_3(0) + t\sigma^1 e^{-iD_5t}\chi_5(0).$$

If the state is prepared on grade 4, then $\xi = 0$. Equation (6) is the foundational statement of time. Equation (5) is not to be cited as the dynamics after the neighbours have been eliminated.

The kernel K is built from unitary groups. The even Gaussian does not select a direction. Dimensional flow does not select a direction. The projected entropy of a pure full state oscillates when the hidden space is two neighbouring grades. An arrow requires an extra reduction: a traced bath, a stationary reference state, and a semigroup written for $t \geq 0$. That reduction is optional open-system analysis. It is not part of the boxed operator.

6 Infrared isolation

Hops conserve spatial momentum. If every nearby symbol is massless and $W_N = 0$, then as $k \rightarrow 0$ the hop hybridizes grades 3, 4, 5. Isolated physics on $N = 4$ is then false in the infrared.

The odd profile prevents that collapse provided

$$v \gg t.$$

The same- k self-energy is

$$\Sigma(k, \omega) = \frac{t^2}{\omega - E_3(k)} + \frac{t^2}{\omega - E_5(k)}.$$

Even hop and odd gap cancel the net zero-momentum shift: $\Sigma(0, 0) = 0$. The leading infrared correction is a speed renormalization.

Proposition 2 (Isolated infrared law). *Assume $v \gg t$, a packet prepared on grade 4, $|k| \ll v$, and observation times $t_{\text{obs}} \gg 1/v$. Then*

$$E_4^{\text{eff}}(k) = \left(1 - \frac{t^2}{v^2}\right)|k| + O\left(\frac{t^2}{v^2}|k|^{3/2}, \frac{t^2|k|^2}{v^3}, \frac{t^4|k|}{v^4}\right).$$

At this order the generator is already time-local. Residual memory in (6) is a transient of duration $1/v$.

Golden-Rule rates obtained by tracing the neighbours as baths,

$$\Gamma_\alpha(\omega) = 2\pi t^2 \nu_\alpha(\omega) n_\alpha(\omega),$$

describe a different reduction. They are not the same- k resolvent.

7 Dimensional flow

Sharing one undeformed copy of $\mathbb{R}^3$ across every grade freezes the spatial spectral dimension at three. Putting a higher order $\gamma_N > 1$ only on inner grades makes those slices softer in the heat kernel of the full $\mathcal{D}$. They do not dominate small τ . Grade index and heat-kernel time are different axes. Observed ultraviolet reduction, if it exists, must be written on $N = 4$.

Ansatz 6 (Running order on $N = 4$).

$$\widehat{D_{\text{sp}}(4)}\psi(k) = |k|^{\gamma(|k|)-1}(\sigma \cdot k) \hat{\psi}(k),$$

with

$$\gamma(0^+) = 1, \quad \gamma(\infty) = \gamma_{\text{UV}}.$$

A concrete interpolation is

$$\gamma(k) = 1 + (\gamma_{\text{UV}} - 1) \frac{(k/k_*)^2}{1 + (k/k_*)^2}.$$

The scale k_* is independent of σ_0 and of v .

Then

$$d_S^{\text{UV}}(4) = \frac{3}{\gamma_{\text{UV}}}, \quad d_S^{\text{IR}}(4) = 3.$$

The target $\gamma_{\text{UV}} = 3/2$ gives spatial dimensions 2 and 3. It is a target, not a derivation. Counting spacetime as $d_S + 1$ then yields 3 in the ultraviolet and 4 in the infrared. The older slogan of a spacetime dimension approaching two is not this operator.

The projected heat kernel

$$K_4(\tau) = \text{Tr}(\Phi_4 e^{-\tau \mathcal{D}^2} \Phi_4)$$

satisfies, to second order in the isolated-grade expansion,

$$K_4(\tau) = \left(1 - \frac{2t^2}{v^2}\right) K_A(\tau) + O\left(\frac{t^2}{v^2} e^{-\tau v^2}\right) + O(t^4), \quad (7)$$

where K_A is the heat kernel of the interpolating symbol on one copy of $\mathbb{R}^3$. The limits of $d_s(\tau)$ are those of K_A . The leak does not move the crossover set by k_* . Compatibility with isolation further wants $v \gg k_*$, so that neighbour branches sit outside $\tau \sim 1/k_*^2$.

8 Continuum window

The isolated infrared, the time-local generator, and the kinematic leak table hold together only inside

$$t \ll v \gg k_*, \quad |k| \ll v, \quad t_{\text{obs}} \gg 1/v, \quad 0.30 \lesssim \sigma_0 \lesssim 0.40. \quad (8)$$

There are three independent scales: leak width σ_0 , hierarchical gap v , and spatial interpolation k_* . No relation among them is derived.

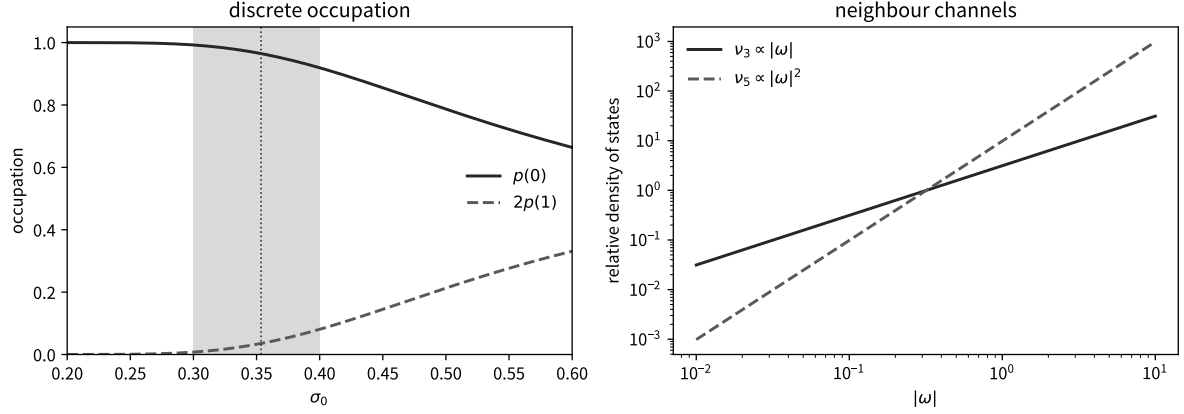


Figure 1: Left: discrete occupation of the unit lattice as a function of σ_0 . The shaded band is the working window $0.30 \lesssim \sigma_0 \lesssim 0.40$; the dashed line is $\sqrt{2}/4$. Right: traced-bath densities of states for the two neighbour channels at the working exponents, $\nu_3(\omega) \propto |\omega|$ and $\nu_5(\omega) \propto |\omega|^2$.

9 Open tasks

1. Derive, or rule out, a unique σ_0 inside the nearest-neighbour window.
2. Compare the unit map with logarithmic and spectral alternatives.
3. Decide whether a $\mathbb{C}^4$ Dirac fibre, or a quaternionic fibre, is needed.
4. Essential self-adjointness of the continuum octonionic expression, if that expression is ever used.
5. Origin of k_* , if any. The exponents of d_s are not moved by the leak.
6. Origin of the gap v . Without $v \gg t$, the infrared is a hybridized cluster.
7. Dynamical selection of $N = 4$, if any.
8. Lorentz invariance as a theorem rather than an infrared symbol.
9. Quantum kinematics (Born rule, composition of subsystems) as a limit of the graded geometry, if that claim is to be made.
10. An arrow, if one is wanted, from extra structure not present in (2).

Phenomenology is not on this list. It is later than items 1–8.

10 Relation to existing programmes

The working operator is assembled from standard pieces. The language of a Dirac organiser and of a spectral action is taken from Connes and Chamseddine; the present operator is not a reconstruction of the Standard Model algebra. The symbol $|k|^{\gamma-1}(\sigma \cdot k)$ is a Fourier-multiplier form of a fractional Weyl operator; what is added is the grade index and the leak. Calcagni's multifractional programme runs dimension with a position-space scale. The present running is

an order $\gamma(|k|)$ on one grade. Open-system formulae used as an optional reduction are Redfield theory, a secular projection, and Spohn's inequality. No new theorem in those subjects is claimed.

11 Conclusion

Hierarchical Shell-Manifold Theory, as a foundation, is now this. Integer-graded immanent shells; a Gaussian residual coupling in a nearest-neighbour window; an associative Dirac-type operator that is essentially self-adjoint; time as the projected law (6); isolated infrared physics on $N = 4$ only when $v \gg t$; dimensional flow on the inhabited grade through $\gamma(|k|)$; three independent scales; no arrow from the boxed operator; no uniqueness theorem; no Standard Model spectrum.

That is narrower than May 2026 and stronger than a list of slogans. It is the official foundational paper until a later derivation forces a revision of the boxed operator itself.

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